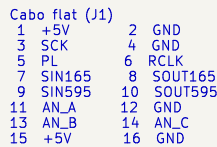


Etiqueta + 16 entradas + 24 saídas para LED. Monte sempre todos os CIs:
a posição de cada byte na cadeia depende disso. Entrada ou saída sem uso fica livre.



Etiqueta: chave k ligada (ON) = bit k-1 igual a 1.
Bits 7-6 = tamanho (médio: 10); bits 5-0 = número da seção (1 a 62).

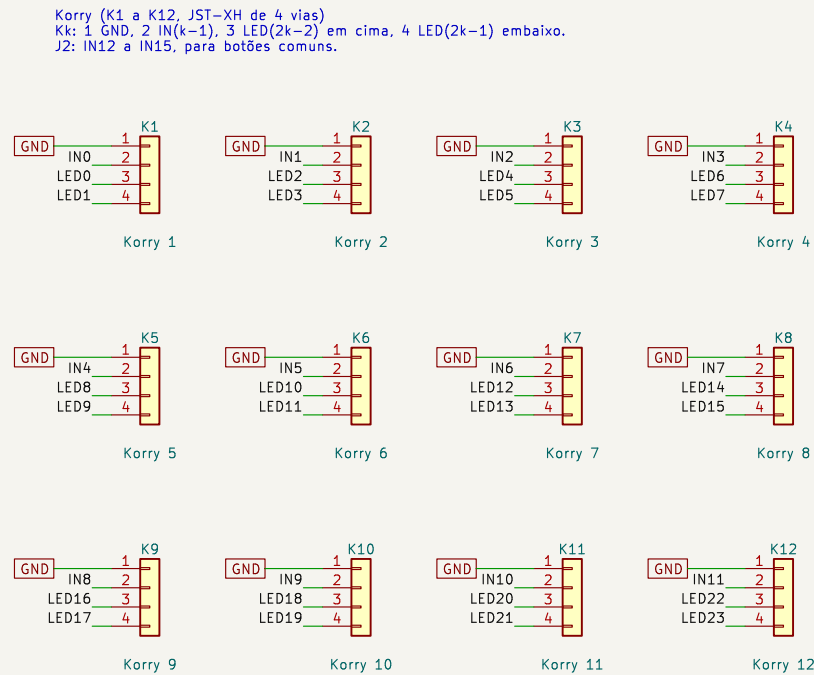
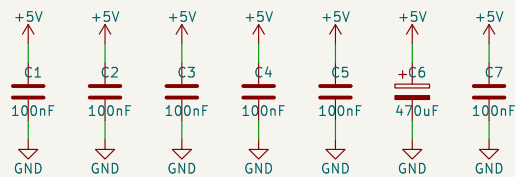
Entradas: SIN165 > U3 > U2 > U1 > SOUT165 (para o Mega).
O Mega lê nesta ordem: U1 (etiqueta), U2 (IN0-7), U3 (IN8-15).
Botão ou chave: entre INn e GND. Apertado/ligado = 0 (pull-up de 10k).

Saídas: SIN595 > U4 > U5 > U6 > SOUT595 (para o próximo slot).
LED: anodo em LEDn, catodo no GND (220: uns 13 mA por LED;
evite os 8 LEDs de um 74HC595 acesos juntos: passa de 70 mA).
Quem acende os LEDs é o Mega, com o estado que vem do jogo:
nenhum LED é ligado direto a um botão.

The image displays three circuit diagrams for connecting a 74HC595 shift register to an Arduino Uno. Each diagram shows the 74HC595 chip with its pins connected to a +5V supply, GND, and Arduino pins.

- Diagram 1 (Left):** Shows the basic connection for 8 outputs (OUT0-OUT7). The 74HC595 chip is connected to a +5V supply (pin 14, SER) and GND (pin 8, GND). The Arduino pins are connected to the 74HC595 pins: SIN595 (pin 14, SER), SCK (pin 11, SRCLK), RCLK (pin 12, RCLK), and GND (pin 13, OE). The 74HC595 outputs are connected to the Arduino pins: OUT0 (pin 15, QA), OUT1 (pin 1, QB), OUT2 (pin 2, QC), OUT3 (pin 3, QD), OUT4 (pin 4, QE), OUT5 (pin 5, QF), OUT6 (pin 6, QG), and OUT7 (pin 7, QH).
- Diagram 2 (Middle):** Shows the connection for 16 outputs (OUT8-OUT15). The 74HC595 chip is connected to a +5V supply (pin 14, SER) and GND (pin 8, GND). The Arduino pins are connected to the 74HC595 pins: CH595_1 (pin 14, SER), SCK (pin 11, SRCLK), RCLK (pin 12, RCLK), and GND (pin 13, OE). The 74HC595 outputs are connected to the Arduino pins: OUT8 (pin 15, QA), OUT9 (pin 1, QB), OUT10 (pin 2, QC), OUT11 (pin 3, QD), OUT12 (pin 4, QE), OUT13 (pin 5, QF), OUT14 (pin 6, QG), and OUT15 (pin 7, QH).
- Diagram 3 (Right):** Shows the connection for 24 outputs (OUT16-OUT23). The 74HC595 chip is connected to a +5V supply (pin 14, SER) and GND (pin 8, GND). The Arduino pins are connected to the 74HC595 pins: CH595_2 (pin 14, SER), SCK (pin 11, SRCLK), RCLK (pin 12, RCLK), and GND (pin 13, OE). The 74HC595 outputs are connected to the Arduino pins: OUT16 (pin 15, QA), OUT17 (pin 1, QB), OUT18 (pin 2, QC), OUT19 (pin 3, QD), OUT20 (pin 4, QE), OUT21 (pin 5, QF), OUT22 (pin 6, QG), and OUT23 (pin 7, QH).

Um 100nF colado em cada CI



Liga no backplane por um cabo flat de 16 vias (IDC 2x8)
Etiqueta + 16 entradas (2 x 74HC165) + 24 saídas para LED (3 x 74HC595)

KSP Cockpit

Sheet: /
File: modulo_medio.kicad_sch

Title: Módulo médio do painel

Size: A3	Date: 2026-09-26
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KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4
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